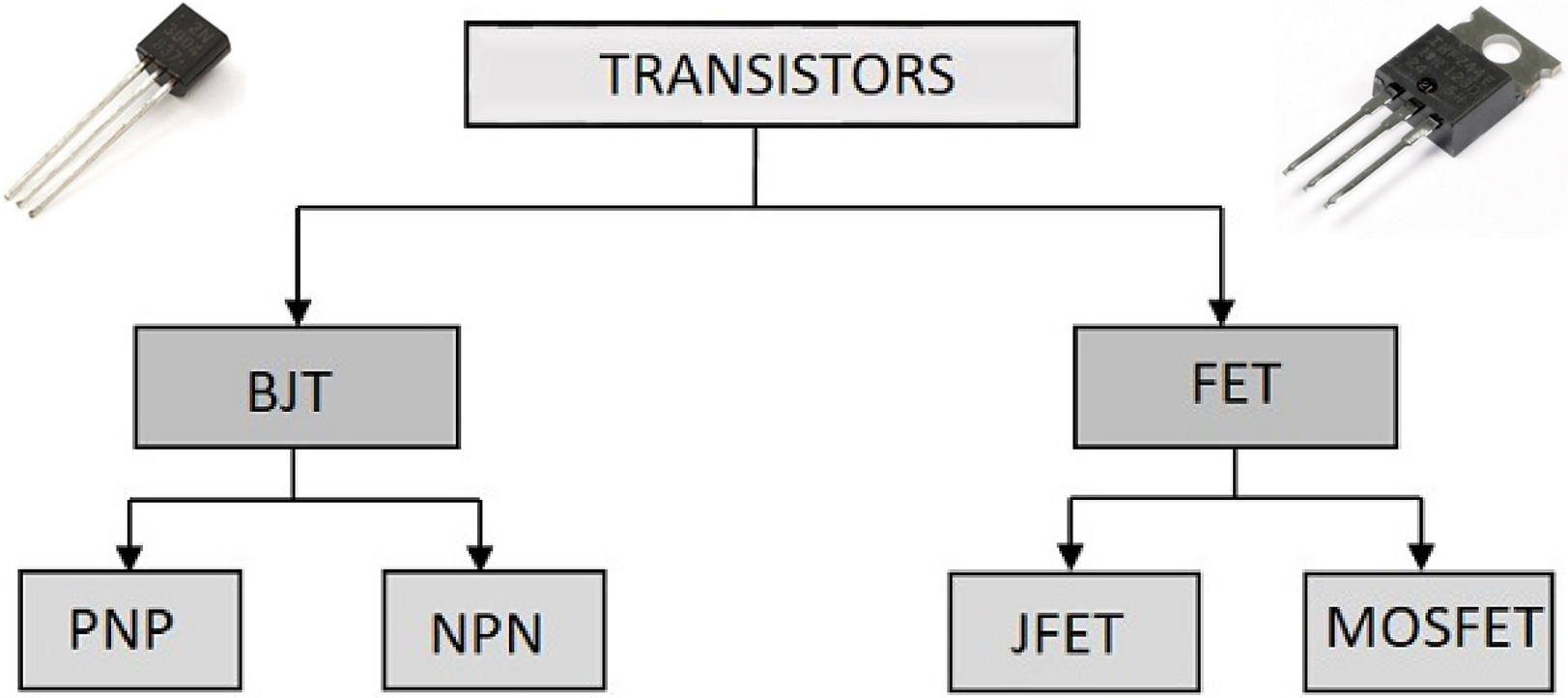
EGCI 232 Engineering Electronics

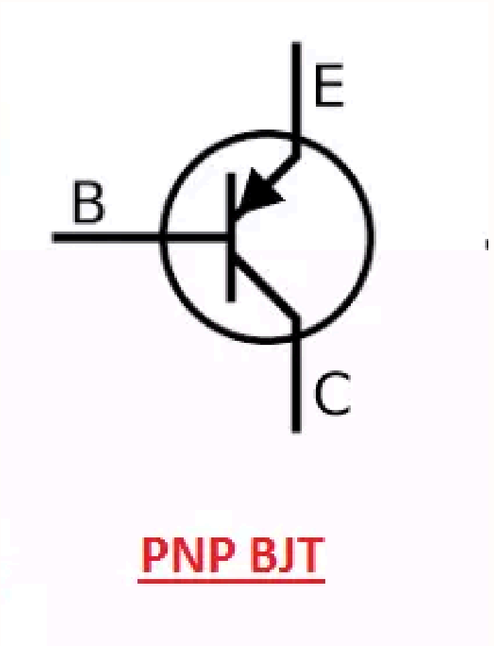
- Bipolar Junction Transistor

Types of Transistors

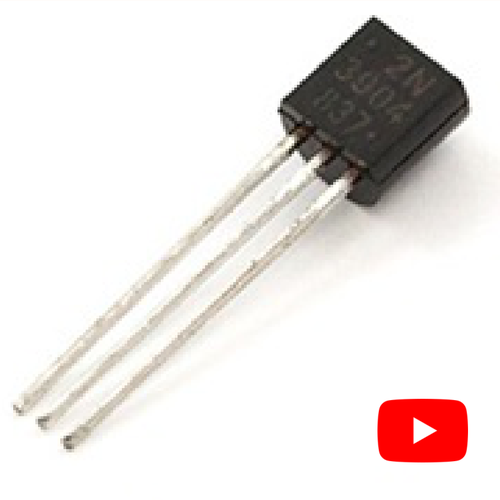


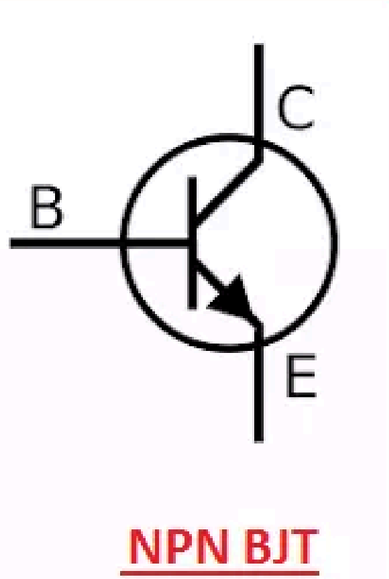


Bipolar Junction Transistors

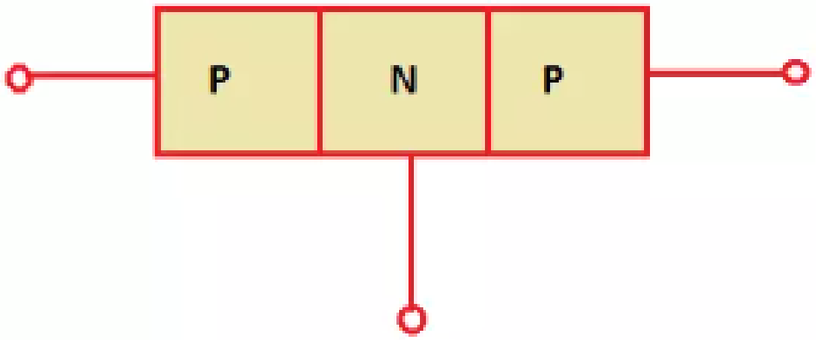


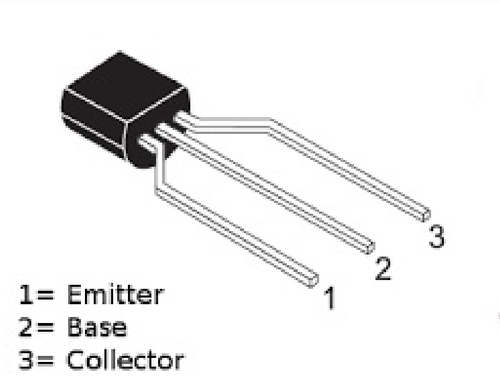
PNP BJT

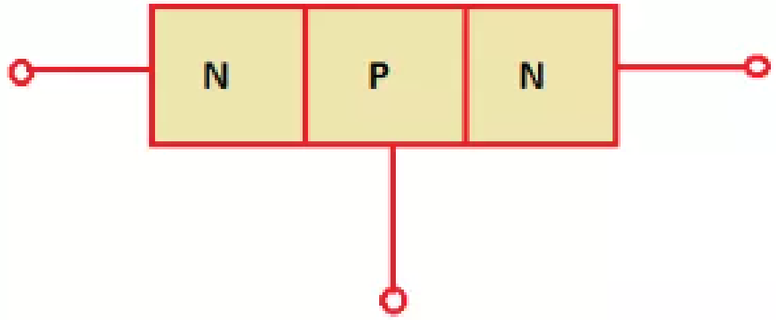




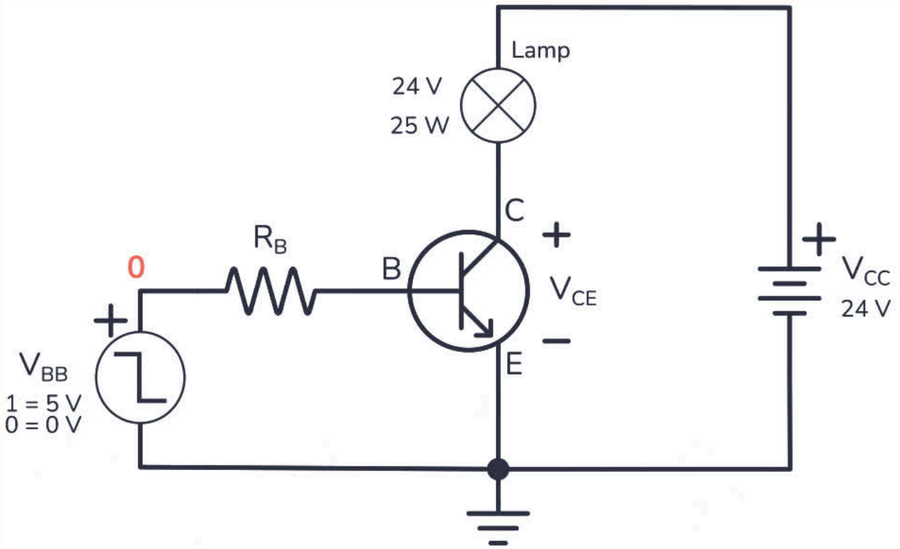
NPN BJT

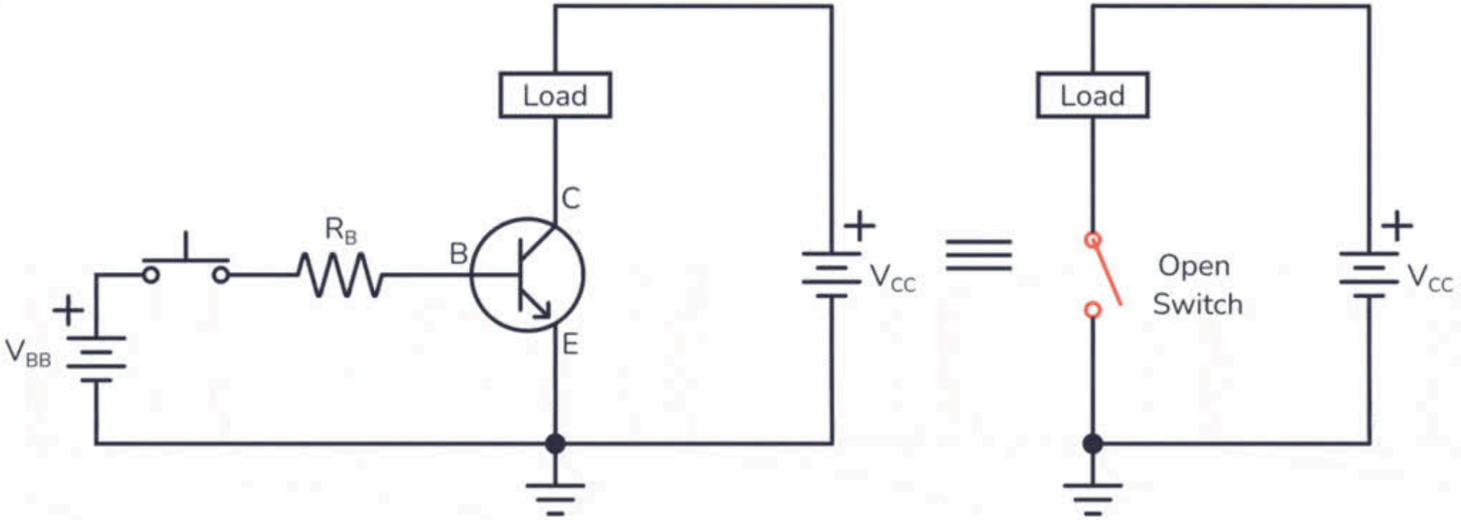






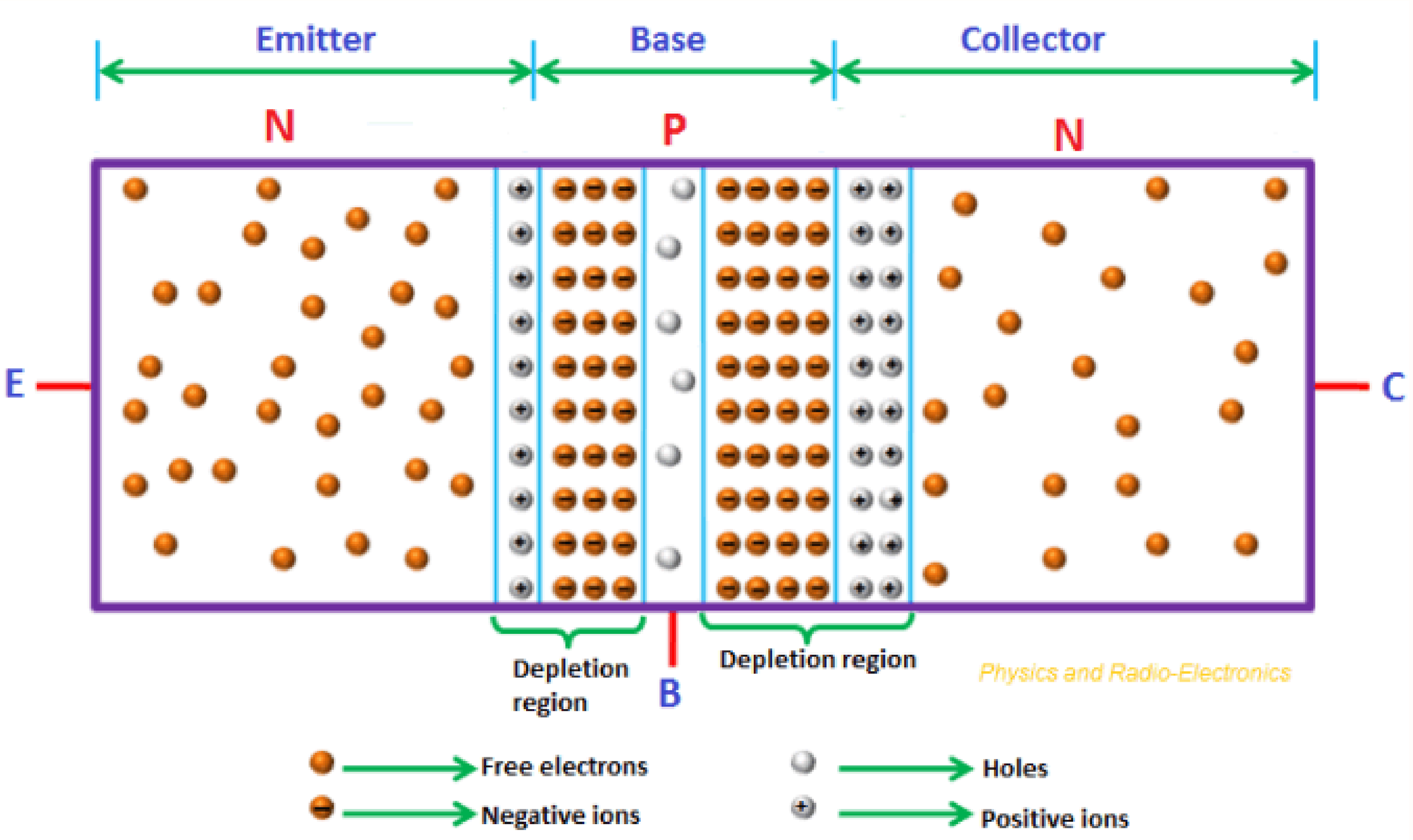
BJT as a Switch



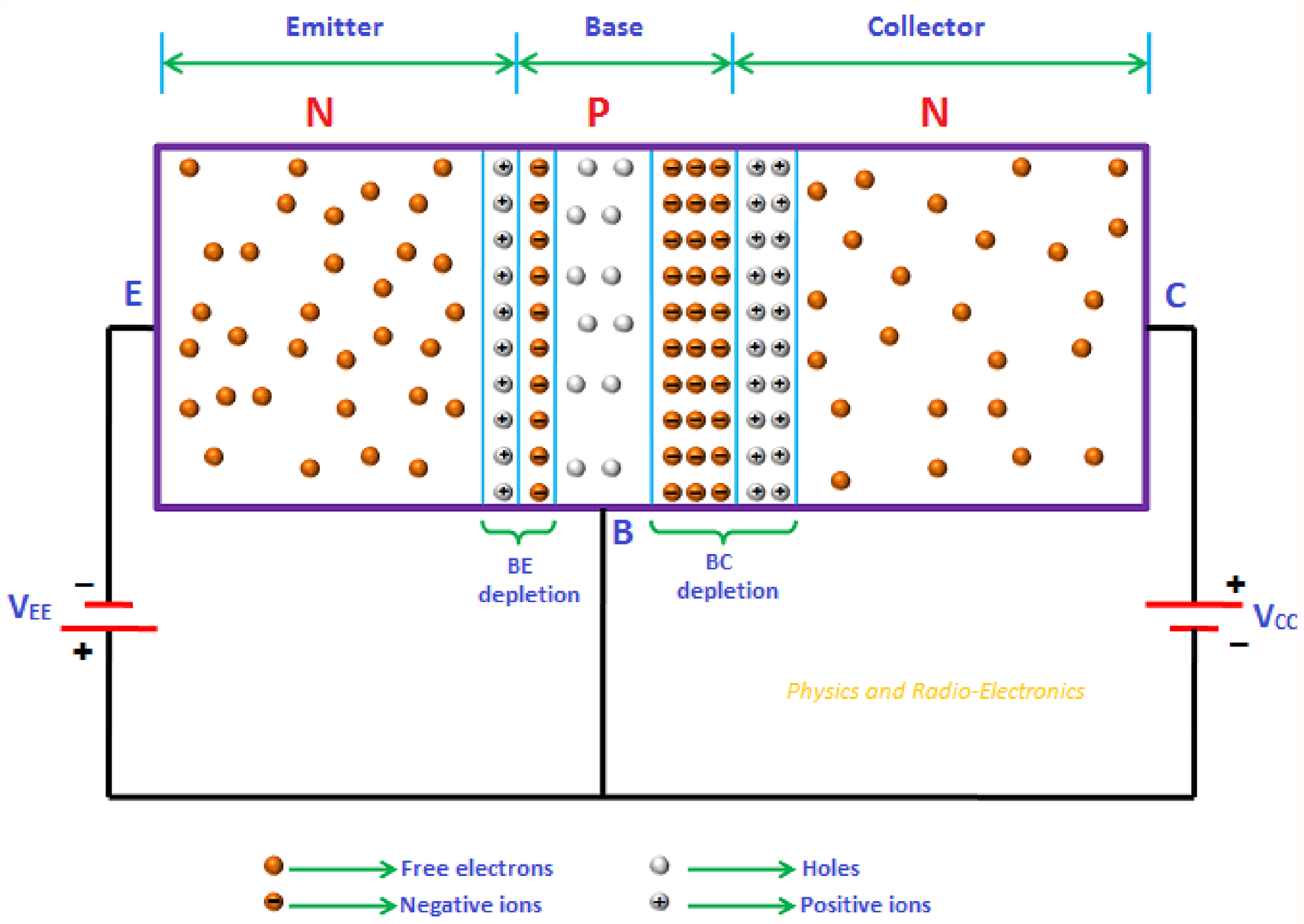


A transistor circuit can easily control things like DC motors, lamps, buzzers, etc.

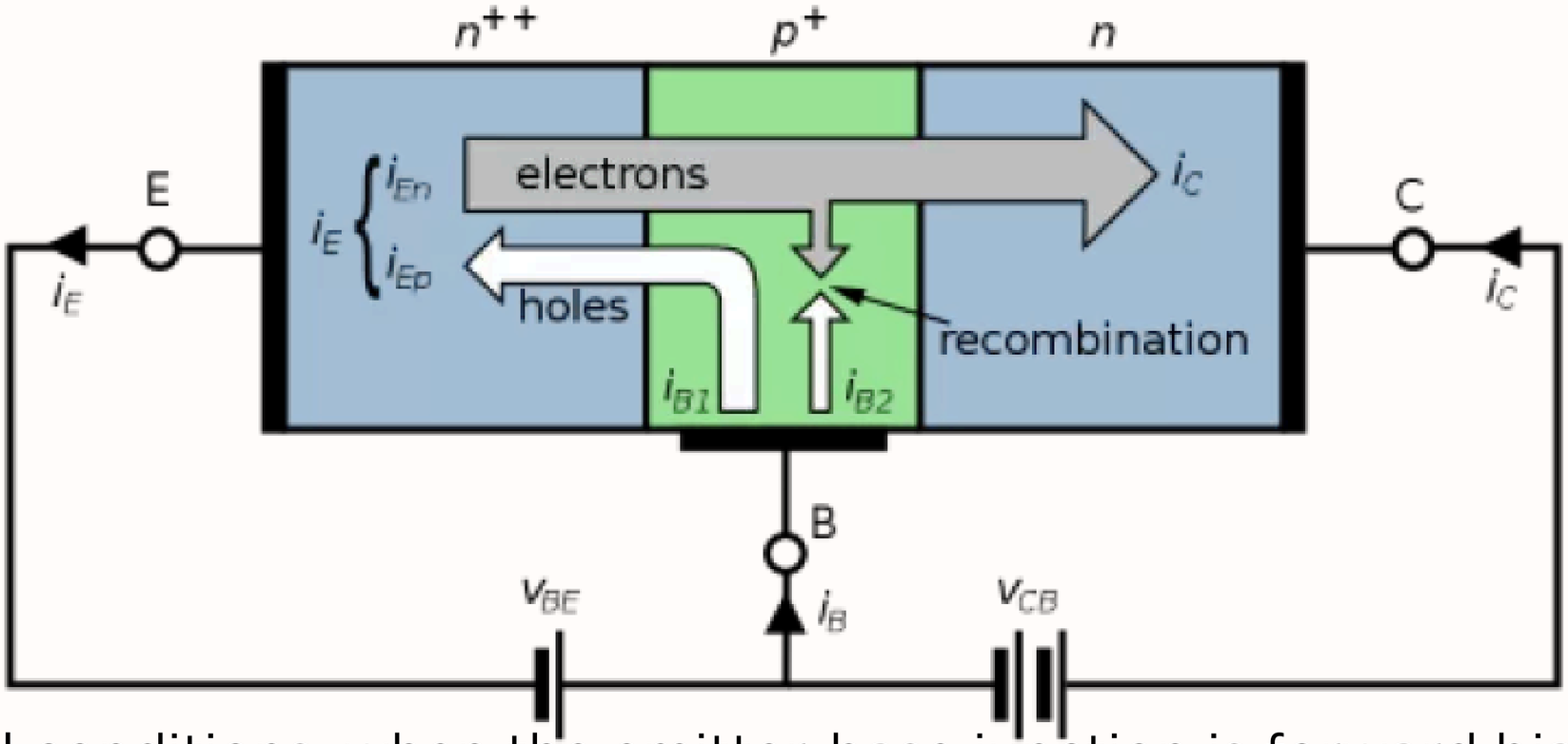
Unbiased NPN Transistor



Biased NPN Transistor

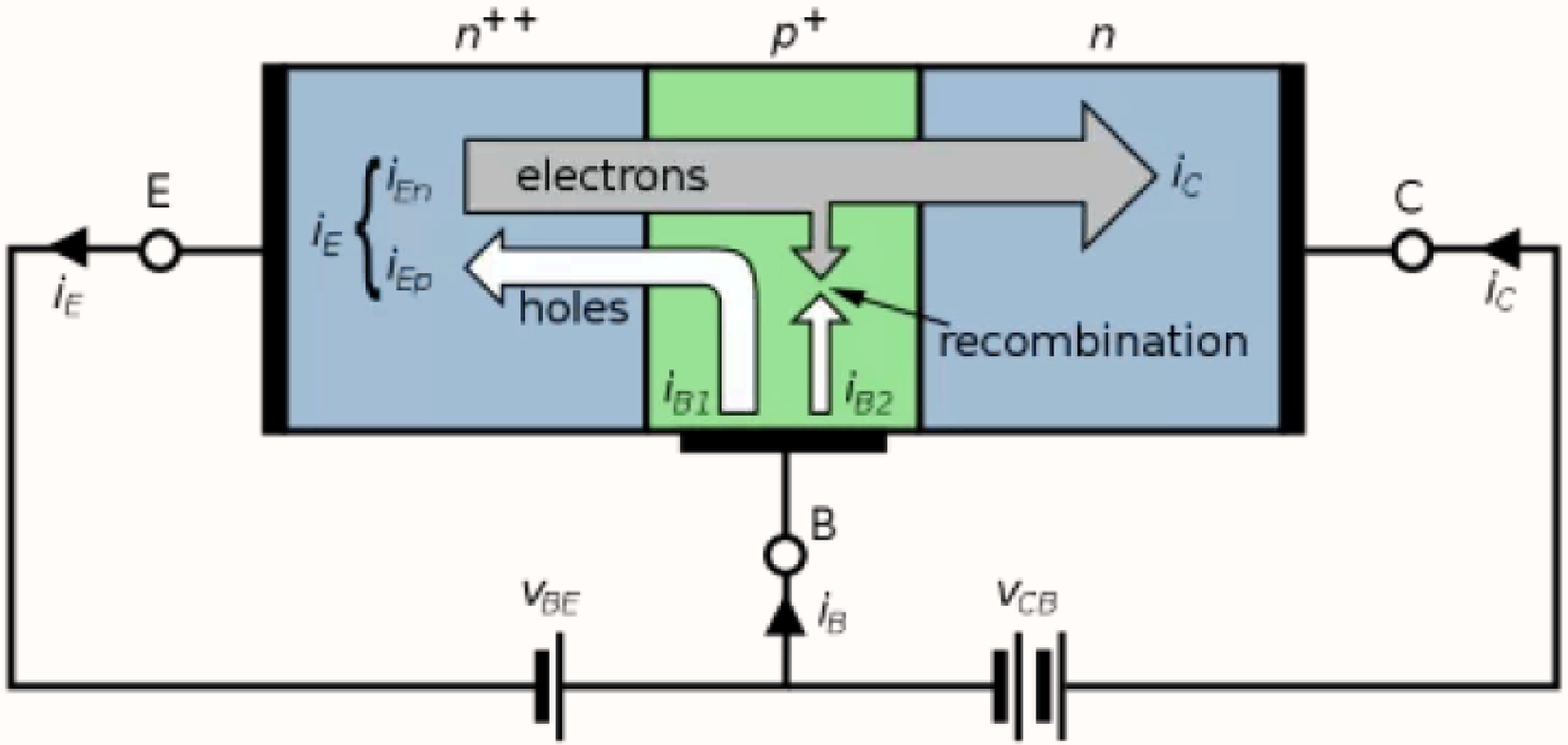


Biased NPN Transistor



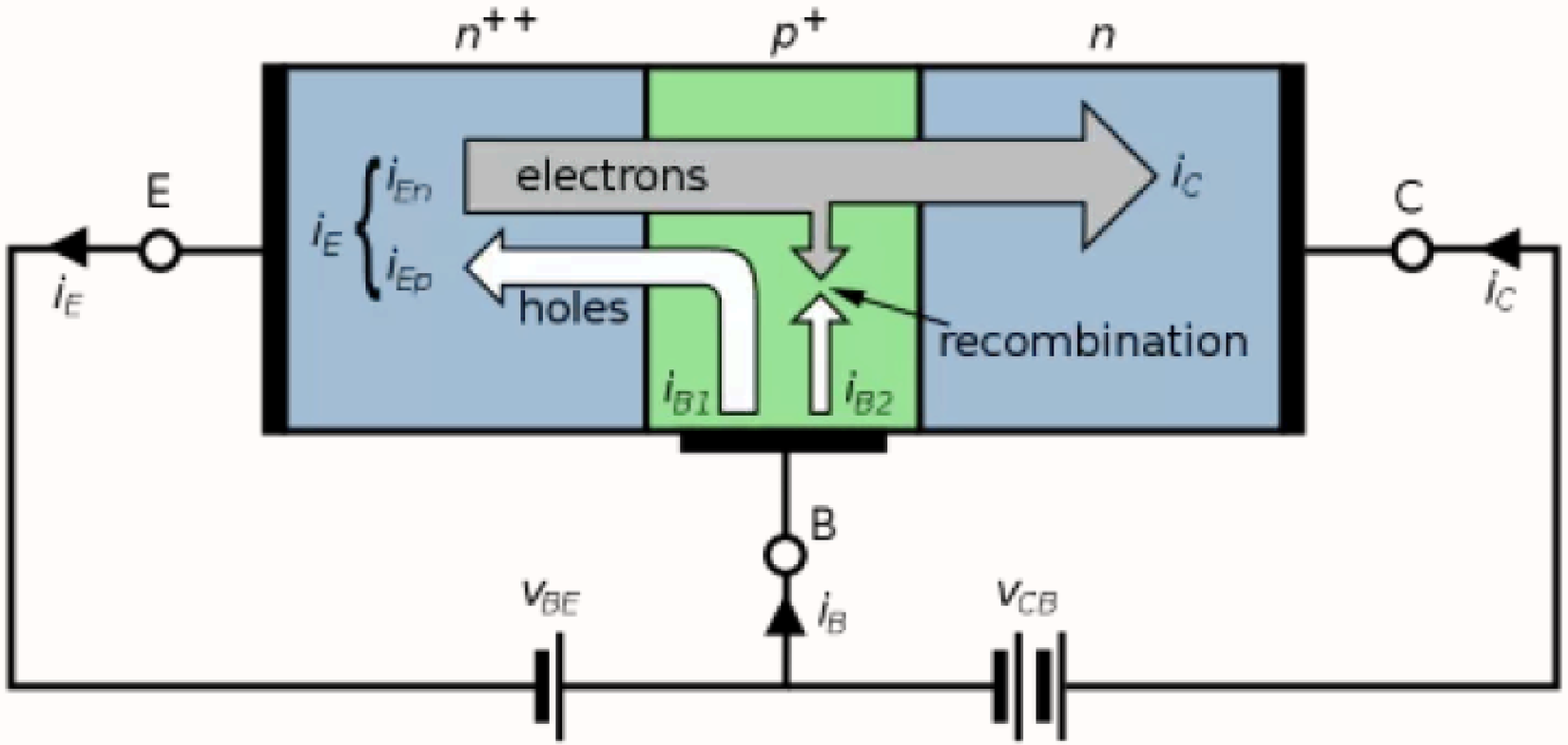
- Under normal conditions, when the emitter-base junction is forward biased it allows the current to flow from emitter to collector.
- When a voltage is applied at the base terminal, it gets biased and draws current, which directly affects the current at the other terminals.

NPN Transistor



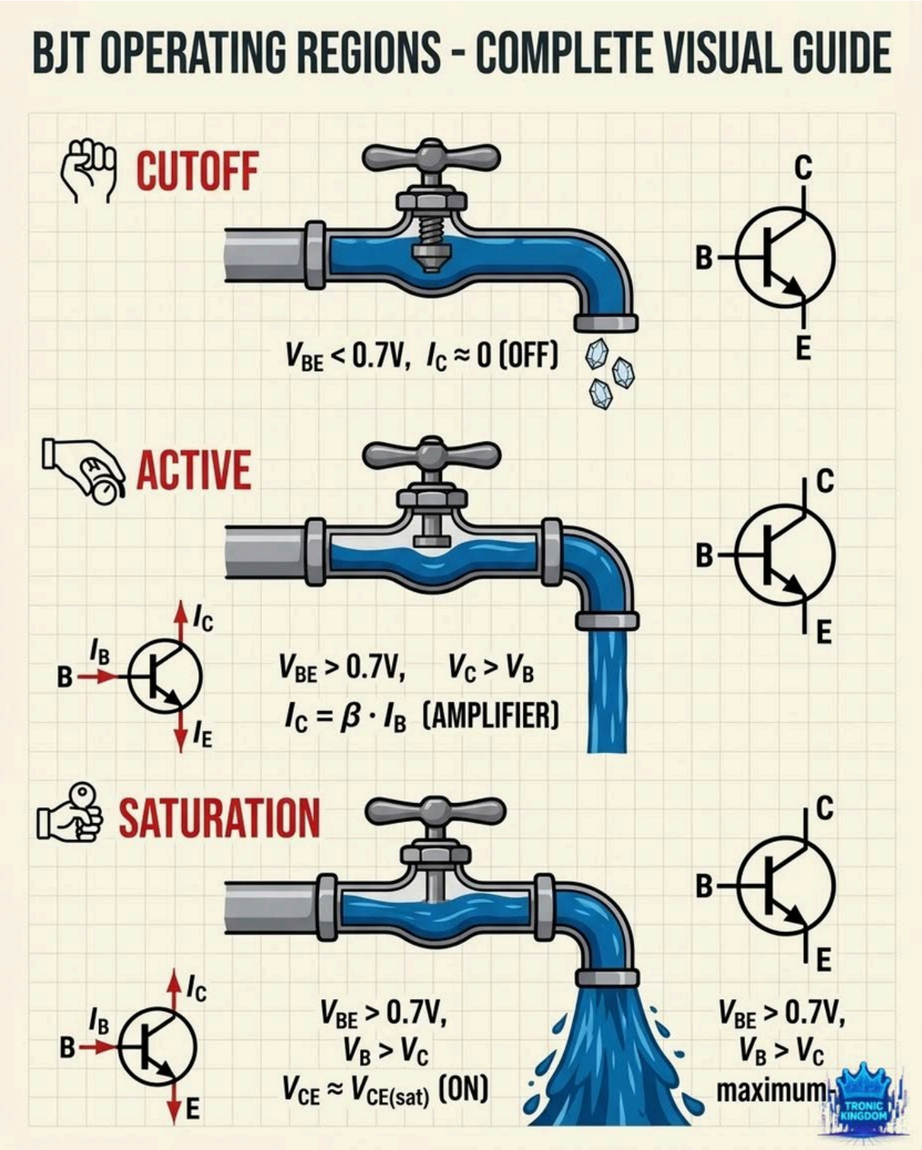
- As the base is very thin and lightly doped it cannot hold the number of electrons for too much time, allowing the electrons to diffuse from base to collector.
- Making a slight change at the voltage applied at the base-emitter terminals can cause a significant change at the current between emitter and collector terminals.

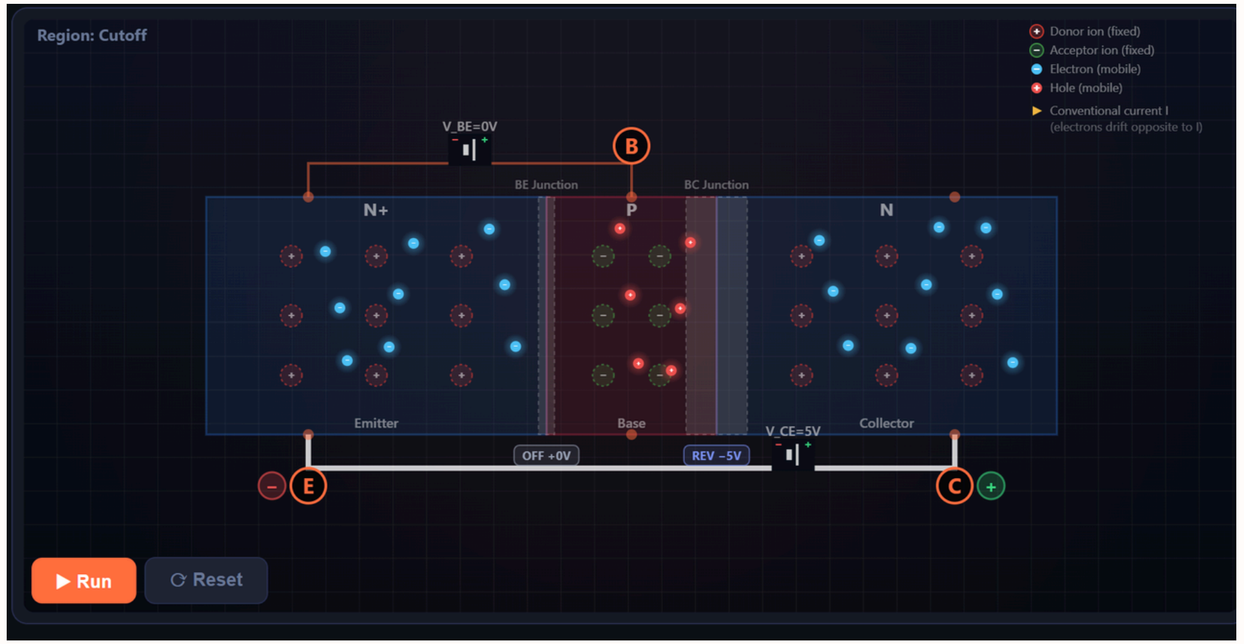
NPN Transistor

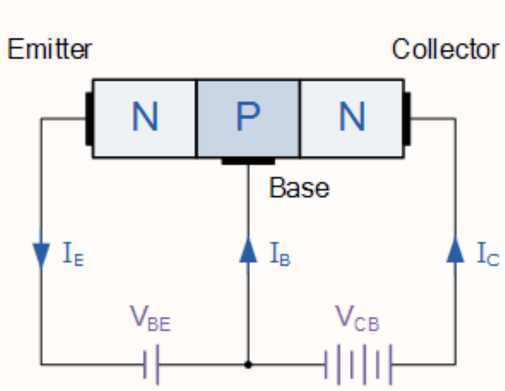


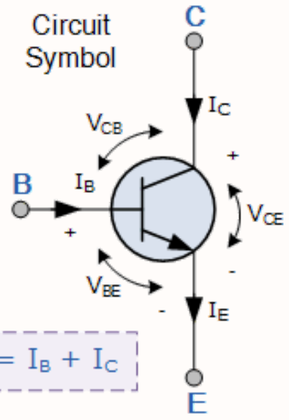
- The mode of operation where an emitter-base junction is forward biased and the collector-base junction is reverse biased is called **active region**.
- This is the process used for **amplification purpose**.

Mode of operations









Mode of operations

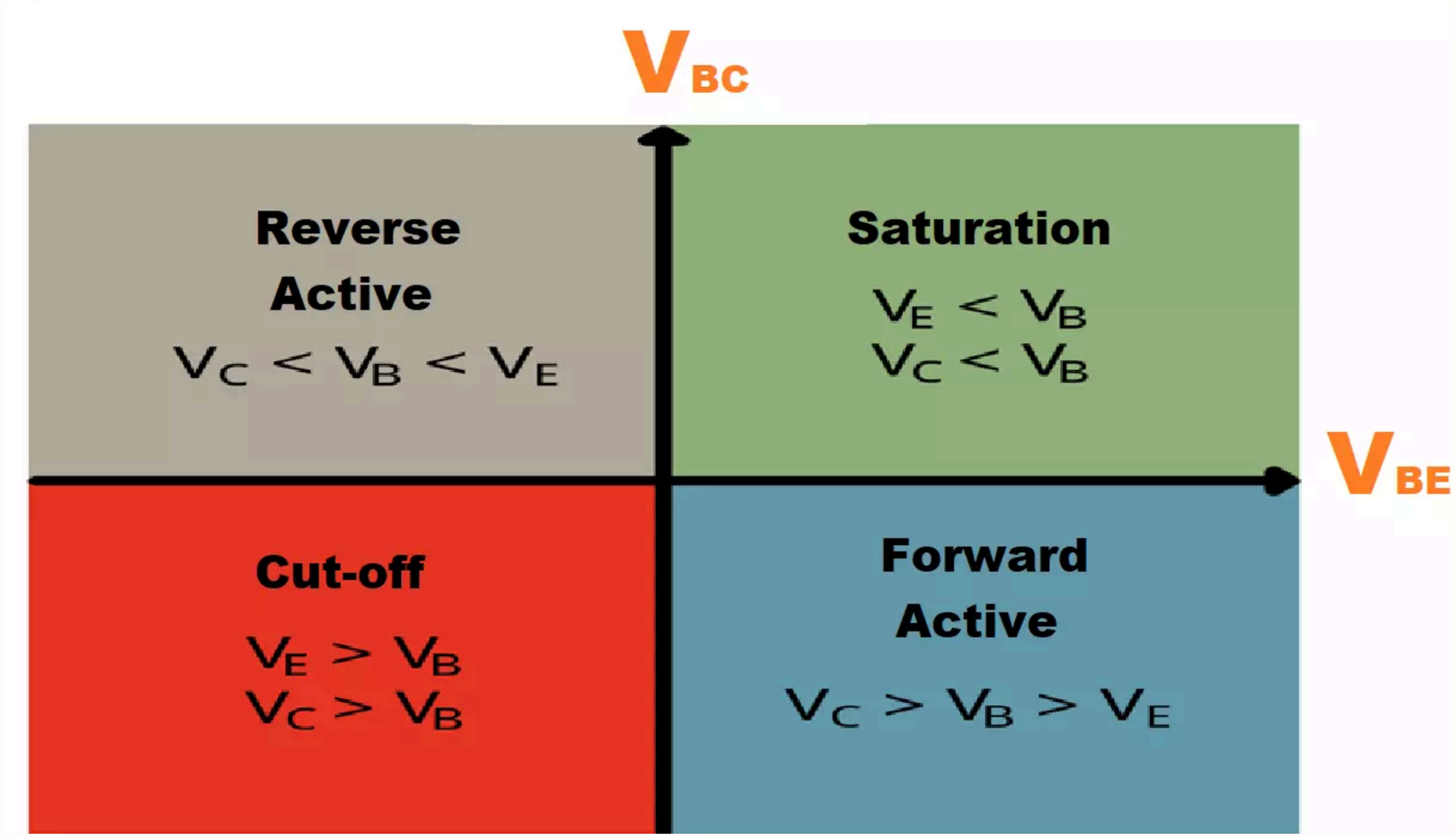
BJT Region of Operation	Emitter-Base Junction	Collector-Base Junction
Active	FB	RB
Cut-off	RB	RB
Saturation	FB	FB
Reverse- Active	RB	FB

Active Region is a region where transistors acts as an amplifier and emitter base junction is forward biased and collector base junction is reverse biased in this region.

Saturation Region is a region where both junctions are forawrd biased and transistor acts as a closed switch or ON switch.

Cut-off Region is considered as fully OFF region where emitter base junction is not forward biased and collector current is zero.

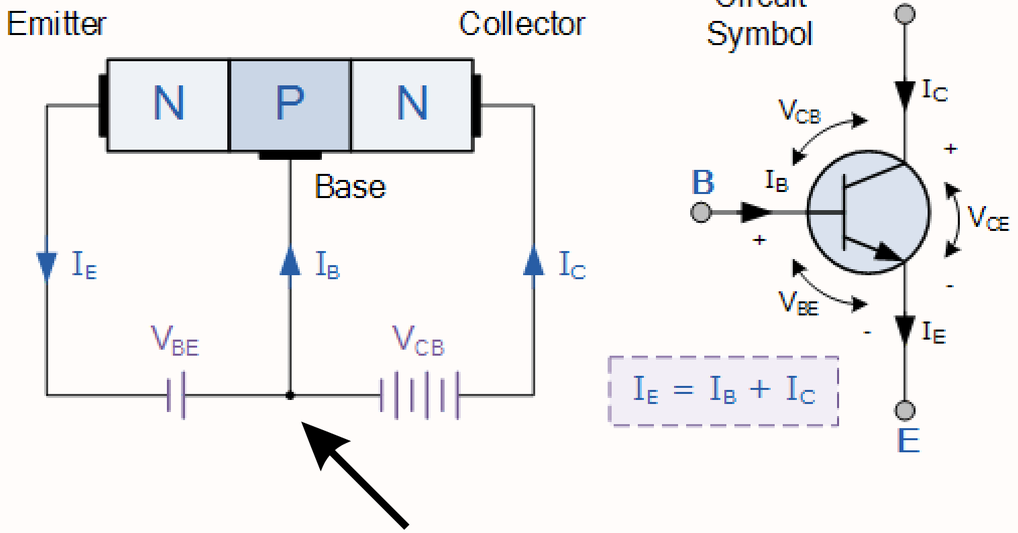
Mode of operations



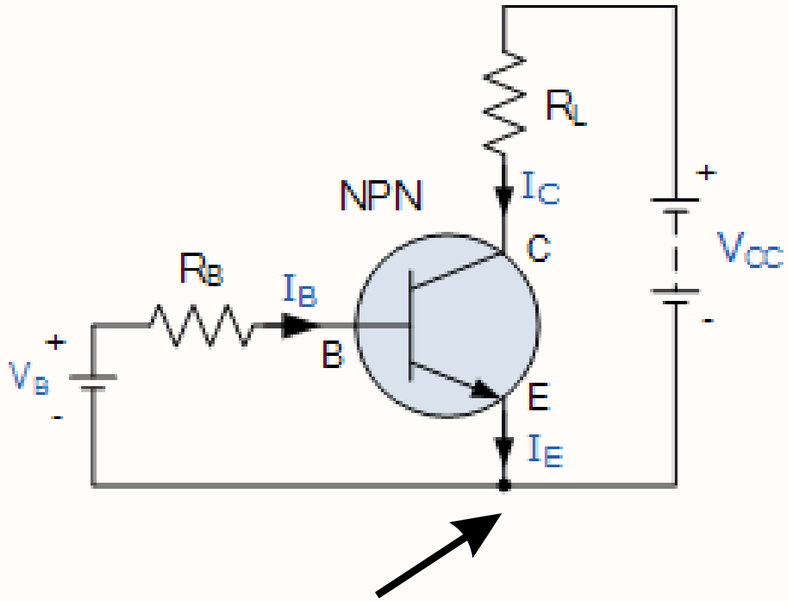
Pros and Cons

PROs	CONs
• Bipolar junction transistor comes with a large amplification factor. • This type of transistor provides a better voltage gain. • This transistor comes with a capability of operating in four regions i.e active region, reverse mode, saturation and cut-off region. • BJT provides a better responese at higer frequiencies. • BJTs also act as a switch.	• BJT is very sensitive to heat and produces noise is some cases. • The switching power of BJTs is very low as compared to unipolar transistors like FETs.

BJT Characteristics

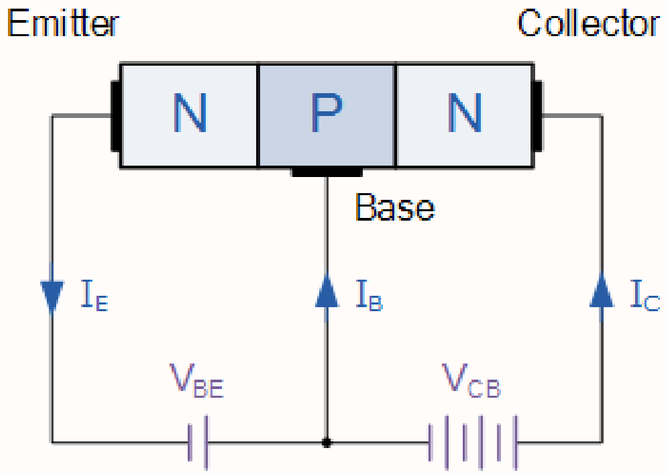


COMMON BASE

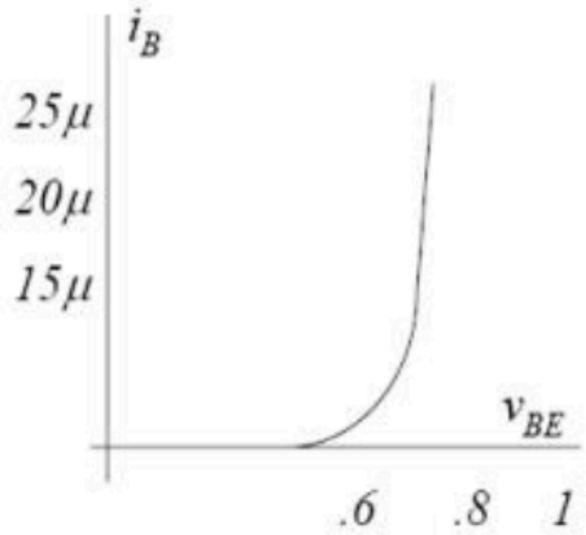


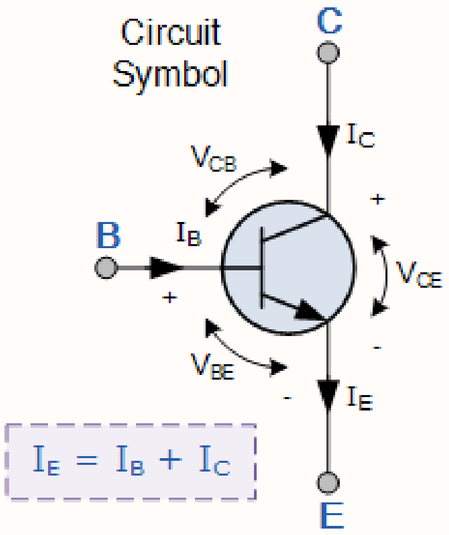
COMMON EMITTER

BJT Characteristics - CB



Emitter-Base Characteristic





$I_E = I_B + I_C$

BJT Characteristics

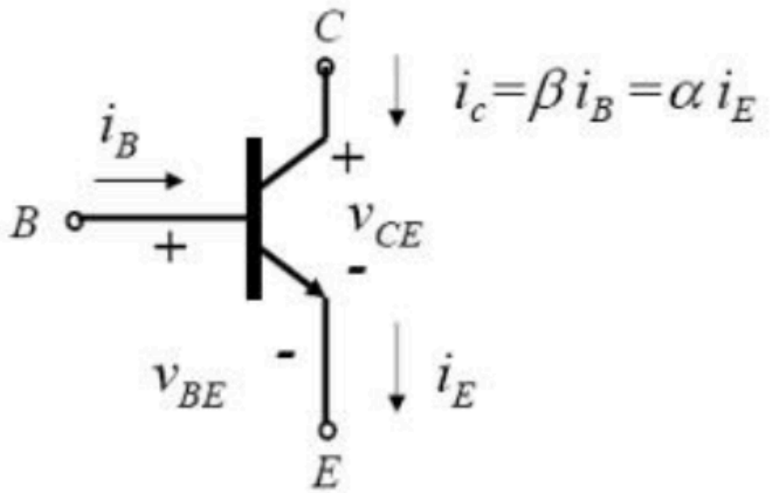
$$i_E = i_C + i_B$$

$$\alpha = \frac{i_C}{i_E}$$

$$i_B = (1-\alpha) i_E$$

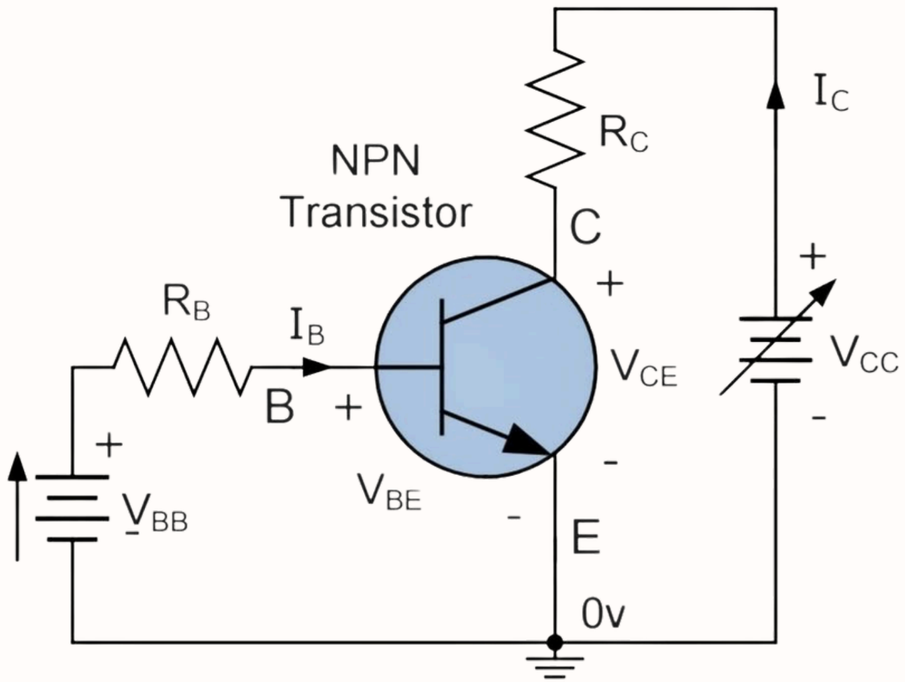
$$i_C = \frac{\alpha}{1-\alpha} i_B = \beta i_B$$

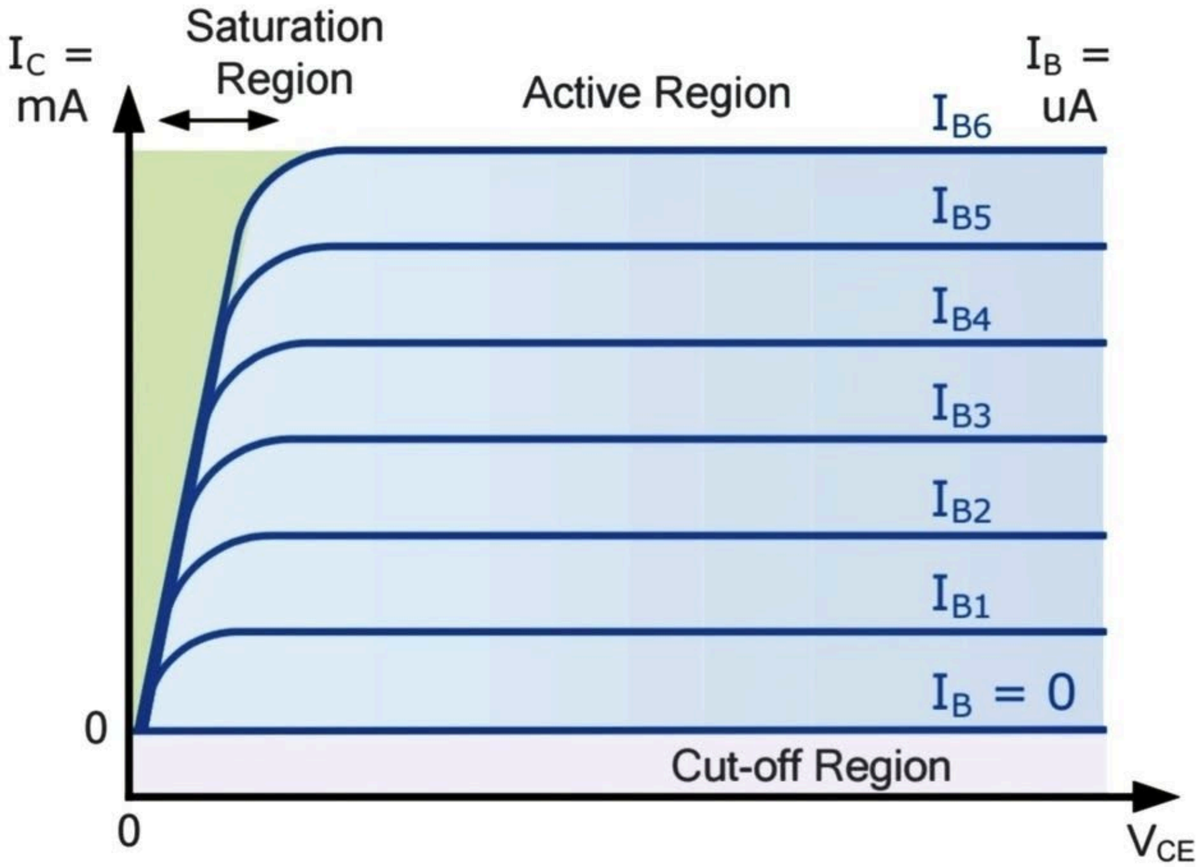
$$\beta = \frac{\alpha}{1-\alpha}$$



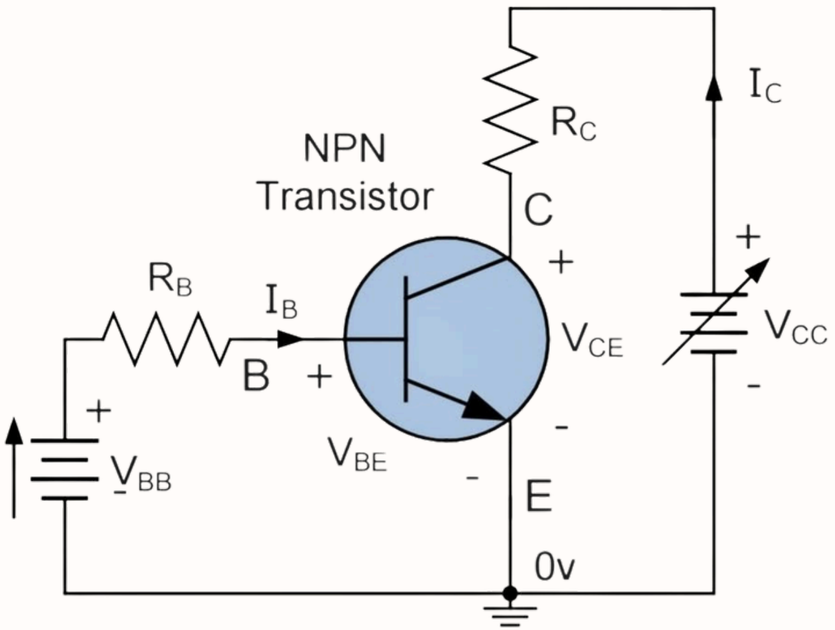
Since α is less than unity then β will be greater than unity and there is current gain from base to collector.

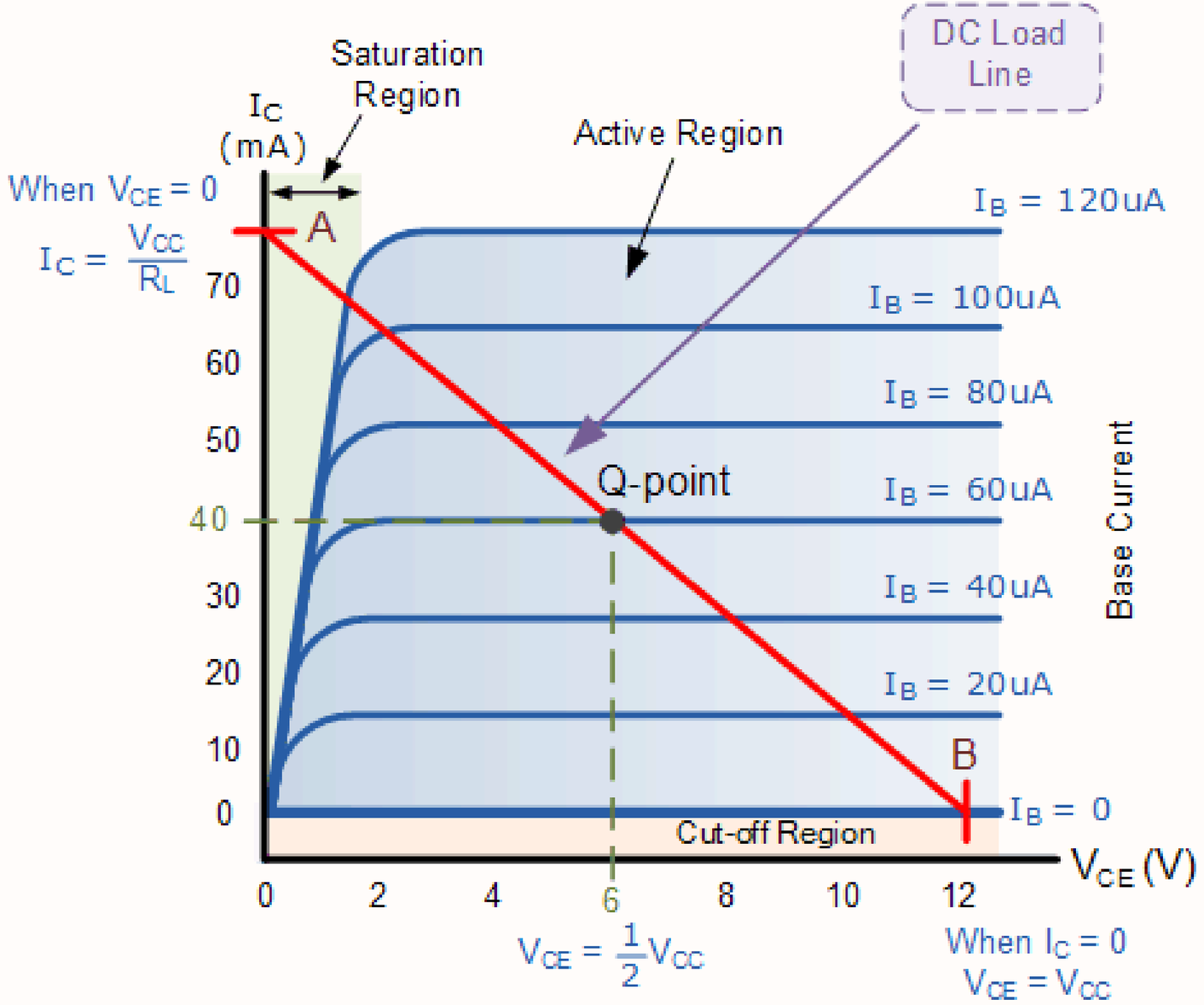
BJT Characteristics - CE



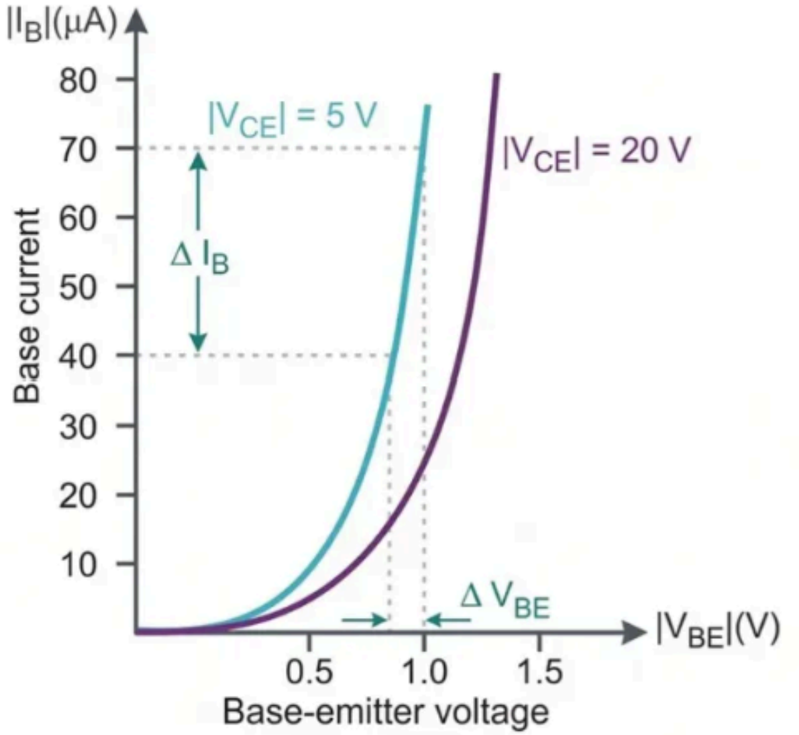


BJT Characteristics - CE

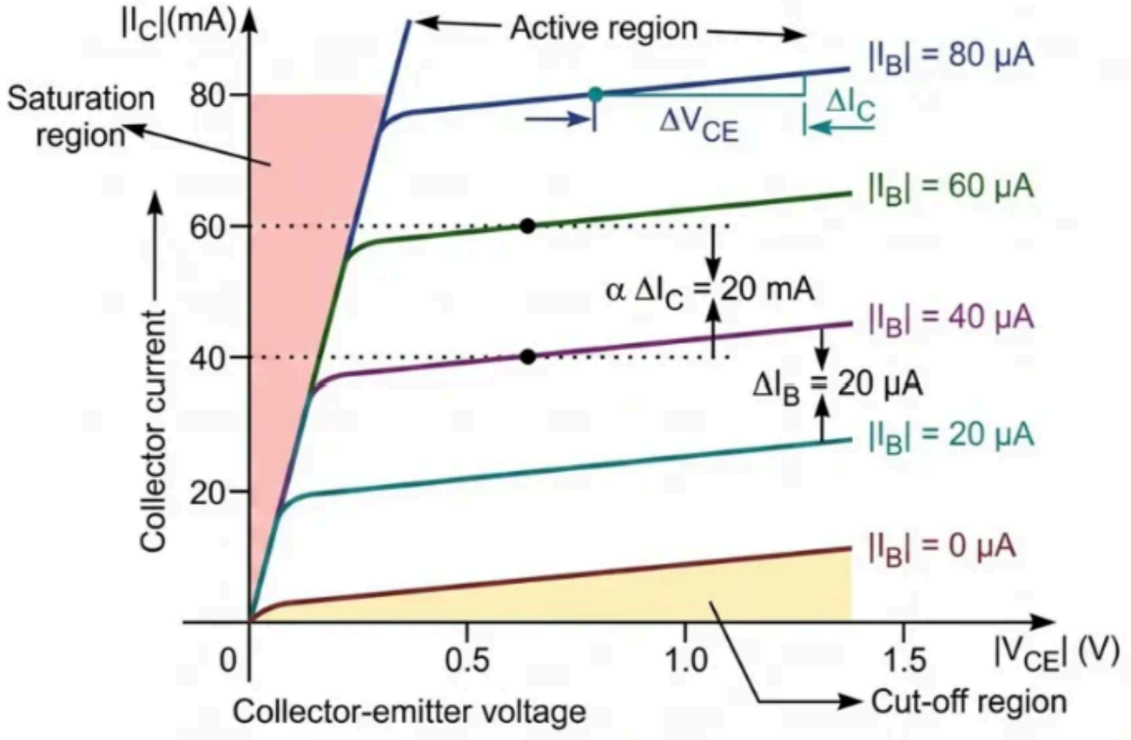




BJT Characteristics - CE

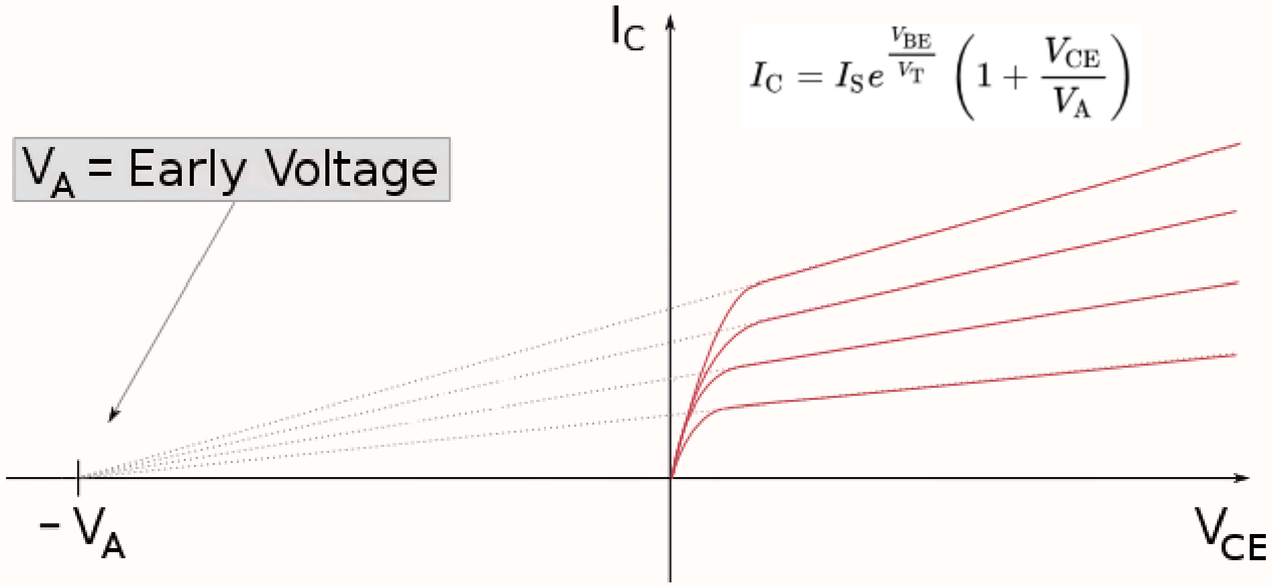


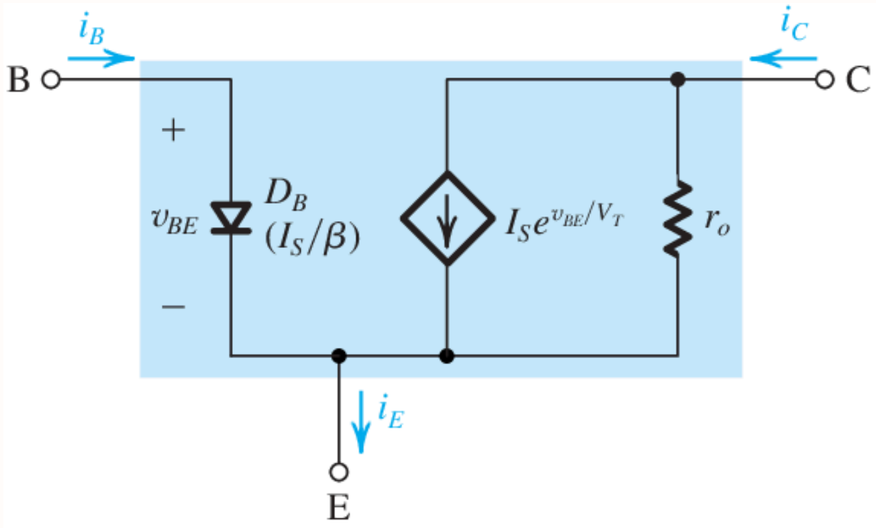
Input Characteristics of BJT in Common Emitter Configurations

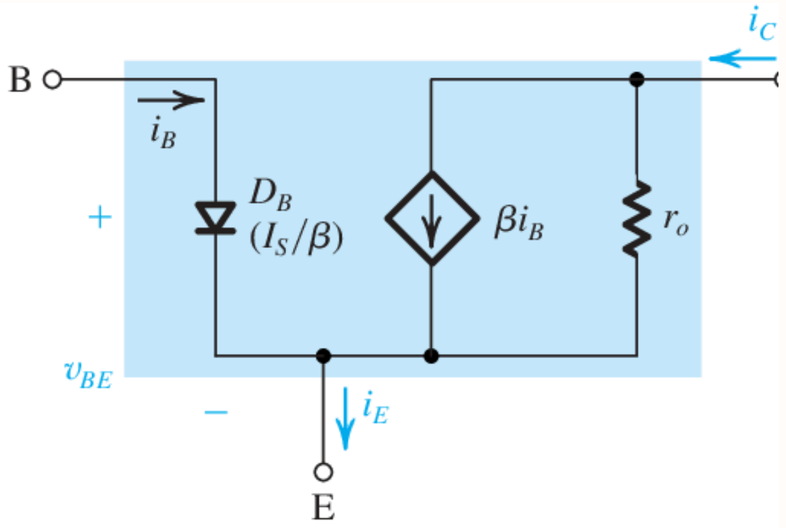


Output Characteristics of BJT in Common Emitter Configurations

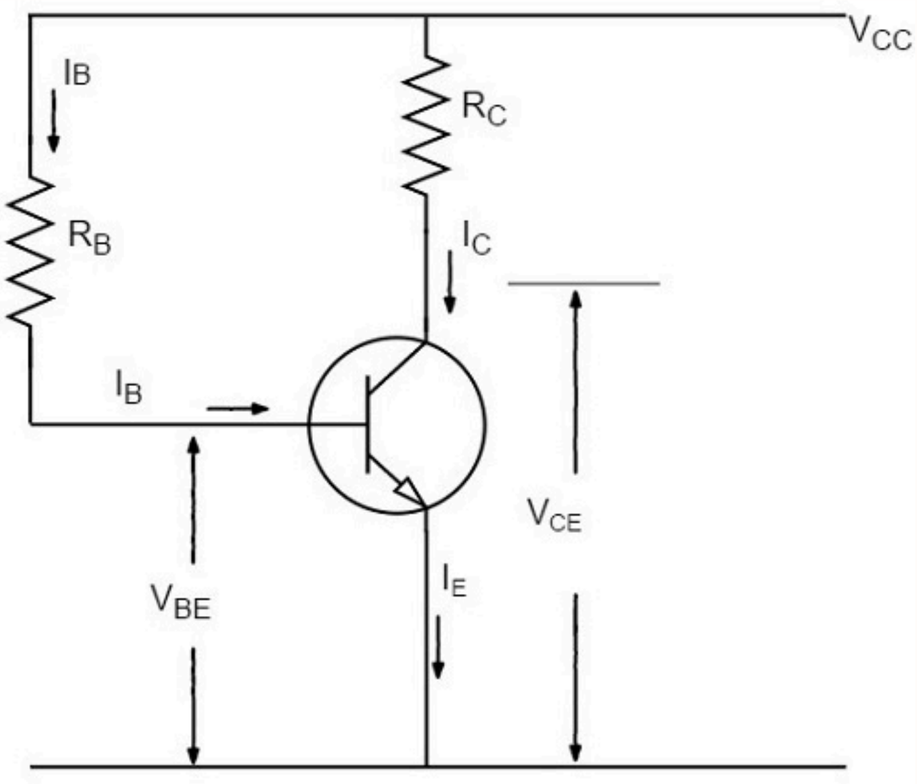
BJT Characteristics - CE



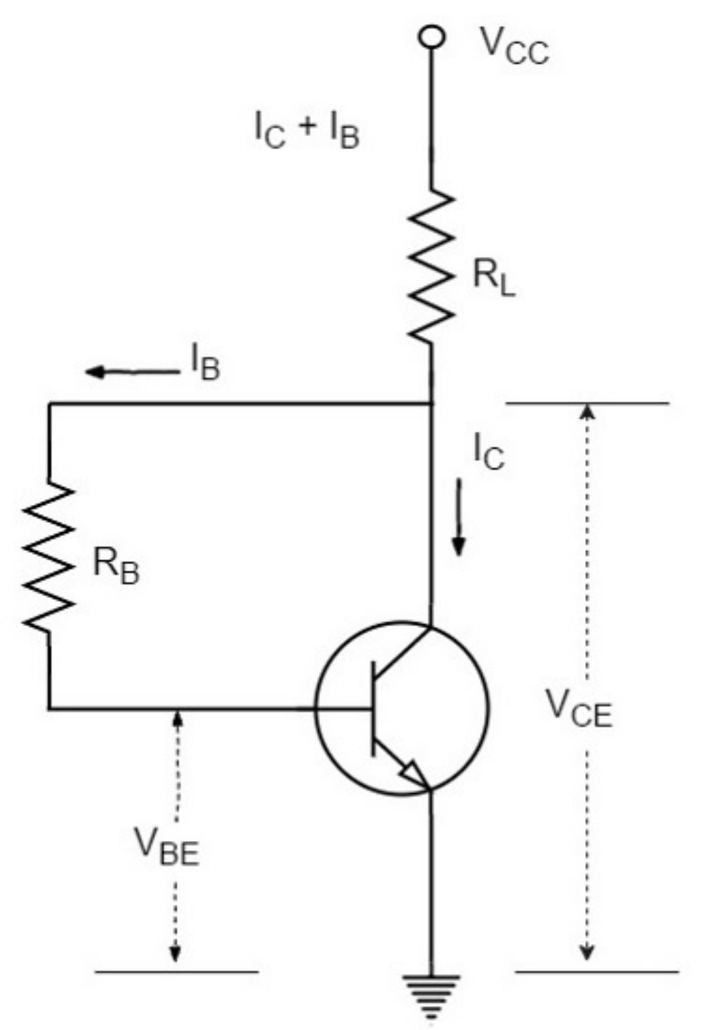




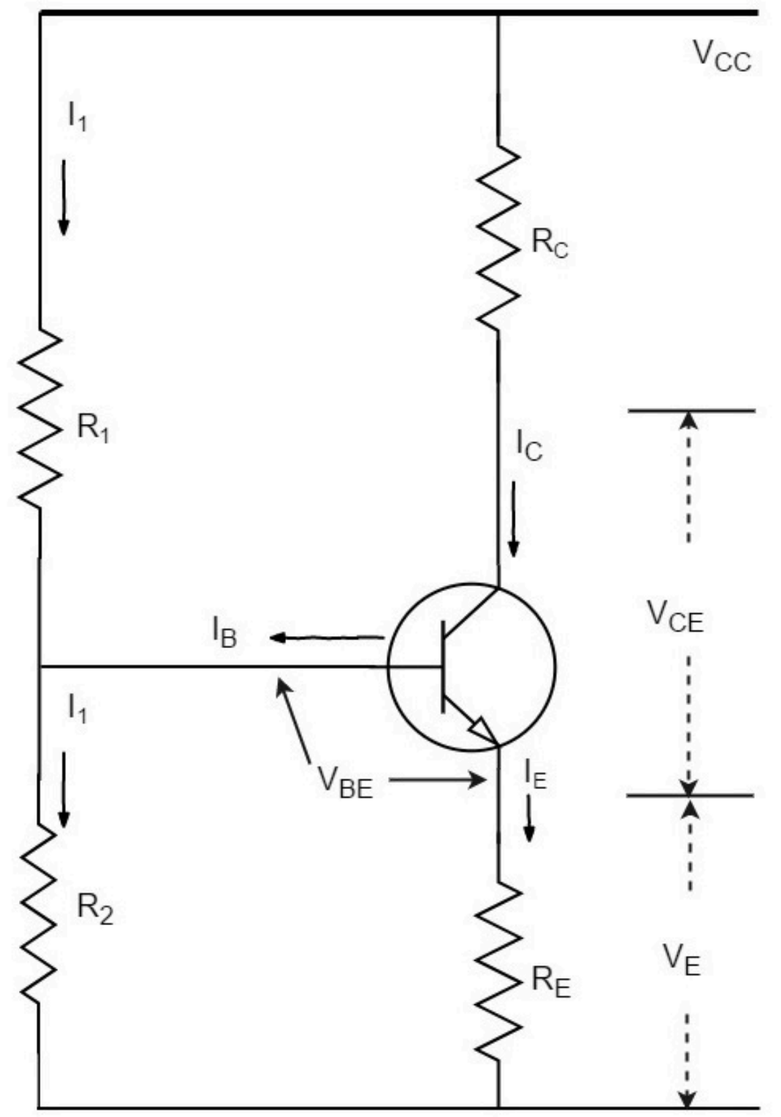
NPN transistor biasing techniques



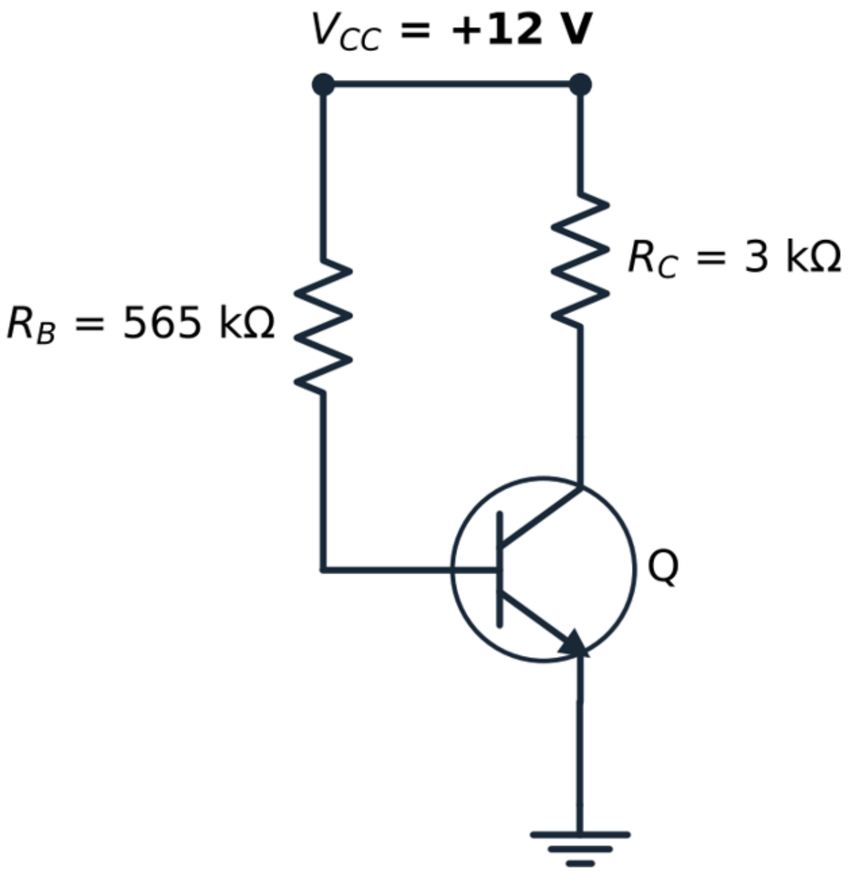
NPN transistor biasing techniques



NPN transistor biasing techniques



Exercise 1: Fixed bias

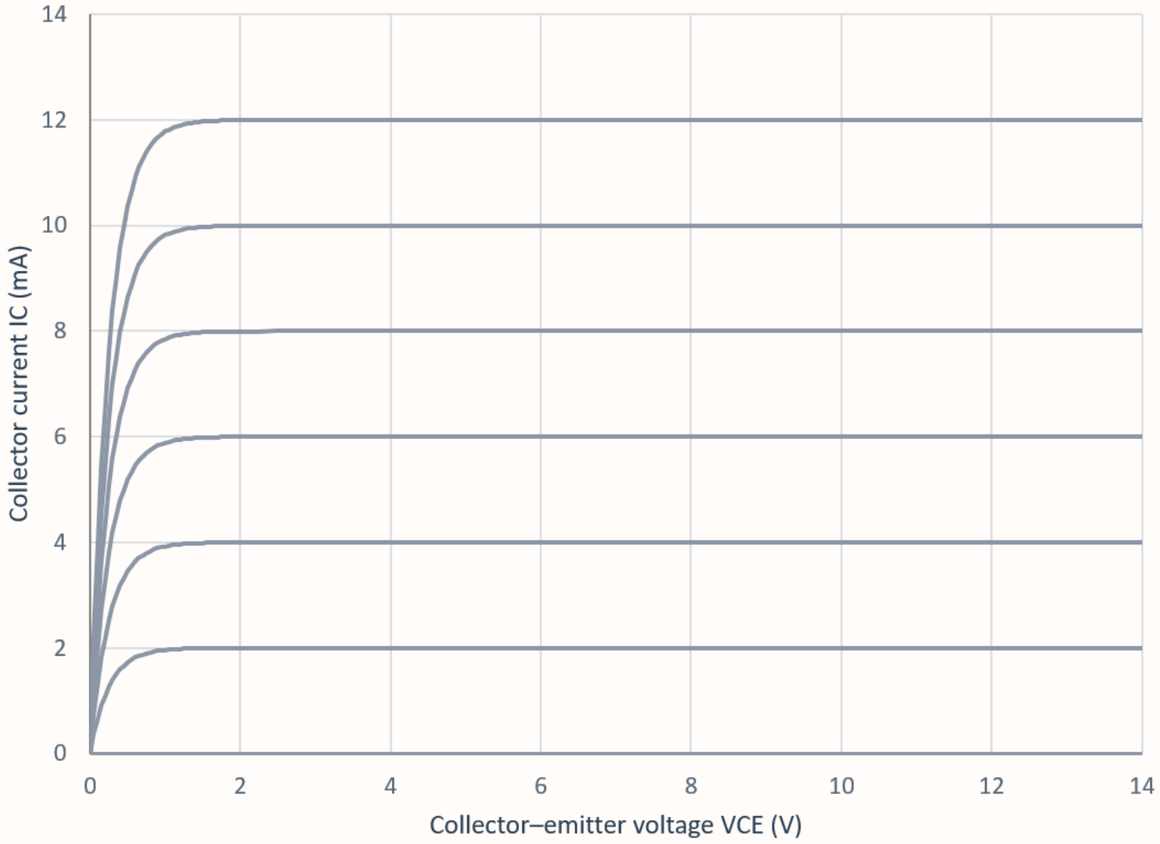


Given

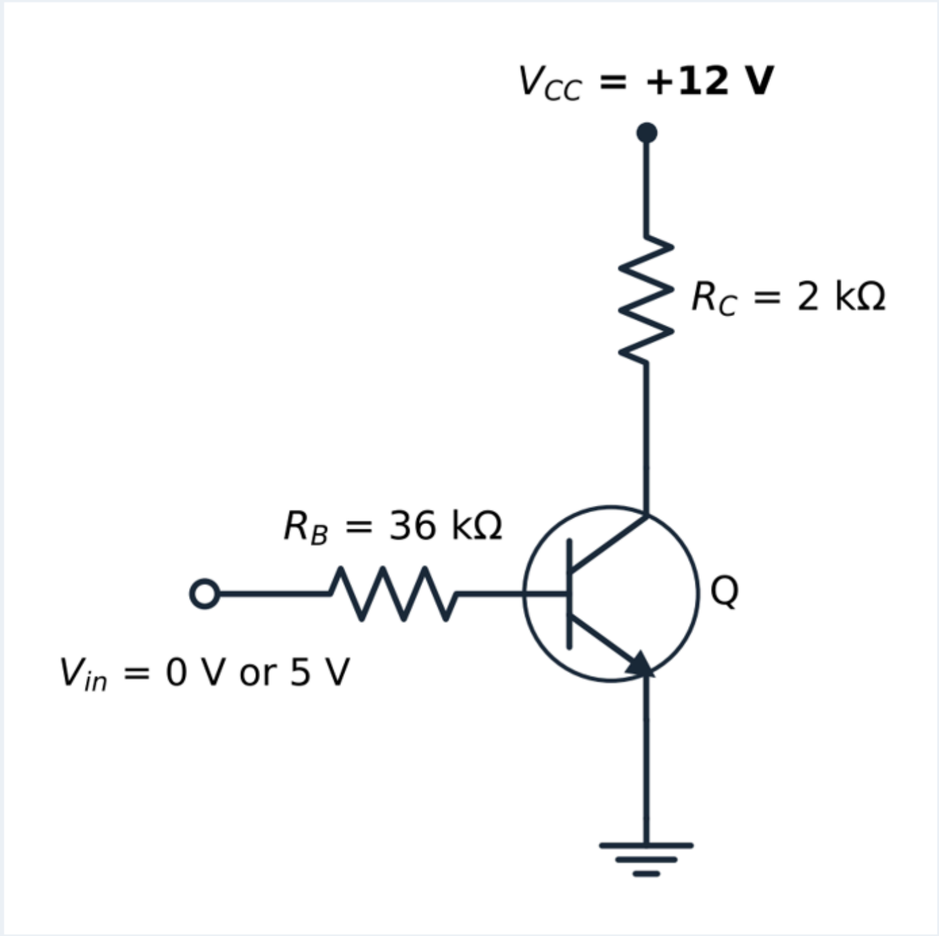
V_{CC} = 12 V
R_B = 565 kΩ
R_C = 3 kΩ
β = 100, V_{BE} = 0.7 V, $V_{CE(sat)}$ = 0.2 V

Find

1. Find I_B from KVL around the base loop.
2. Find I_C and V_{CE} by hand (β = 100).
3. Draw the DC load line on the curve sheet.
4. Read the Q-point. Which region?
5. R_C = 6 kΩ: where is the Q-point now?


14
12
10
8
6
4
2
0
Collector current IC (mA)
0
2
4
6
8
10
12
14
Collector–emitter voltage VCE (V)

Exercise 2: Transistor switch

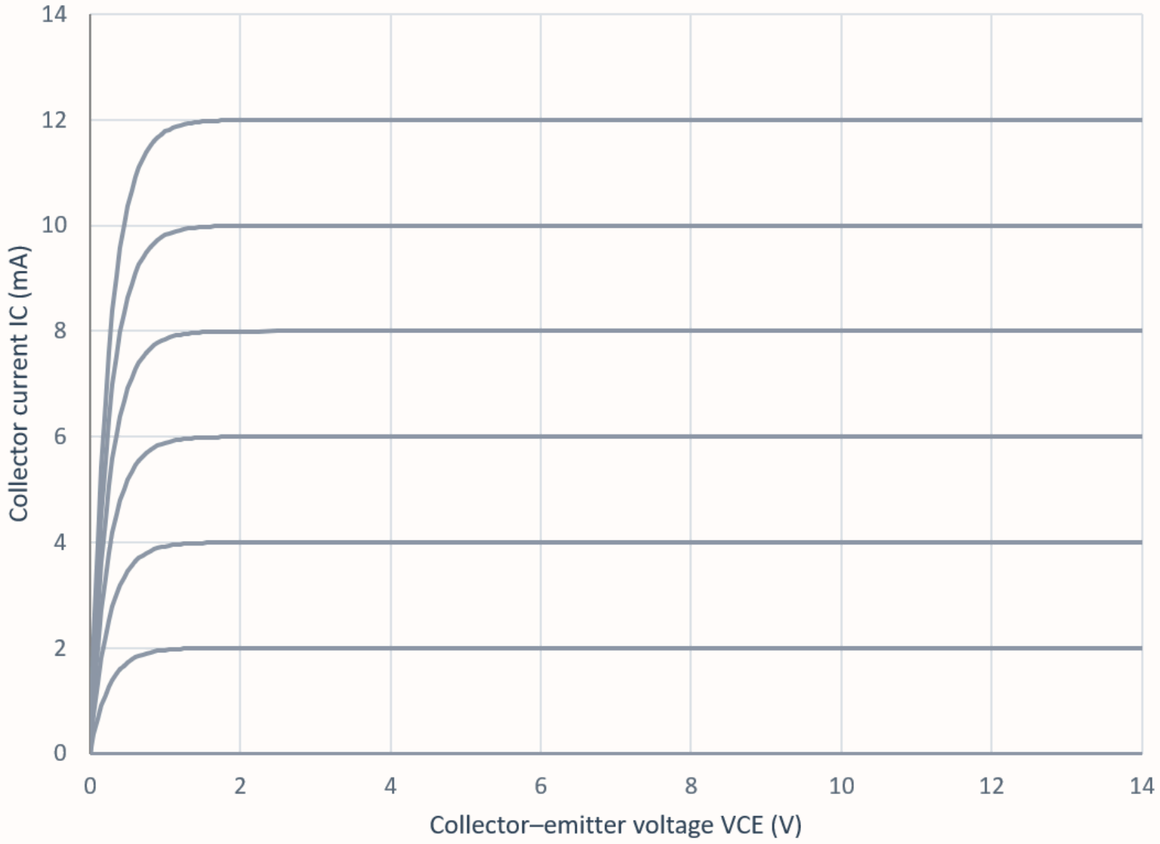


Given

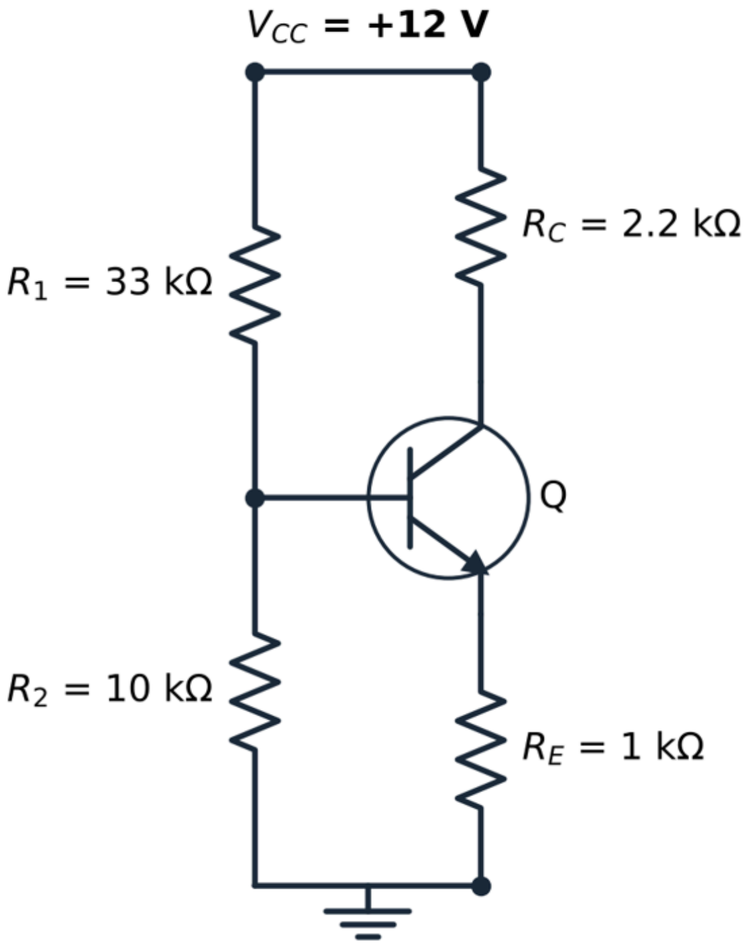
$V_{CC} = 12$ V
$R_B = 36$ kΩ
$R_C = 2$ kΩ
$V_{in} = 0$ V or 5 V
$\beta = 100$, $V_{BE} = 0.7$ V, $V_{CE(sat)} = 0.2$ V

Find

1. $V_{in} = 0$ V: find I_C and V_{CE} (OFF point).
2. $V_{in} = 5$ V: find I_B and $I_{C(sat)}$.
3. Is $\beta \cdot I_B > I_{C(sat)}$? Find the forced β.
4. Mark the ON point on the load line.
5. Largest R_B that still saturates?


14
12
10
8
6
4
2
0
Collector current IC (mA)
0
2
4
6
8
10
12
14
Collector–emitter voltage VCE (V)

Exercise 3: Voltage-divider bias



Given

$V_{CC} = 12\ V$

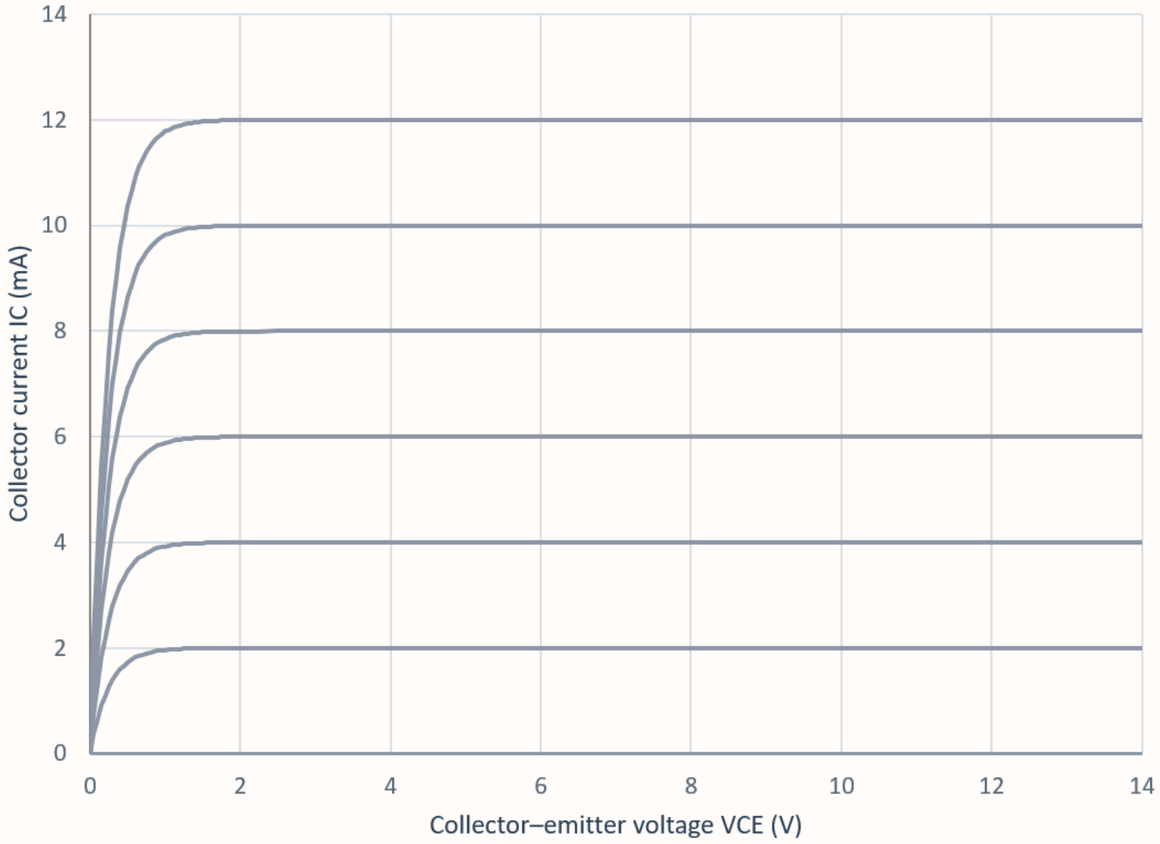
$R_1 = 33\ k\Omega$, $R_2 = 10\ k\Omega$

$R_C = 2.2\ k\Omega$, $R_E = 1\ k\Omega$

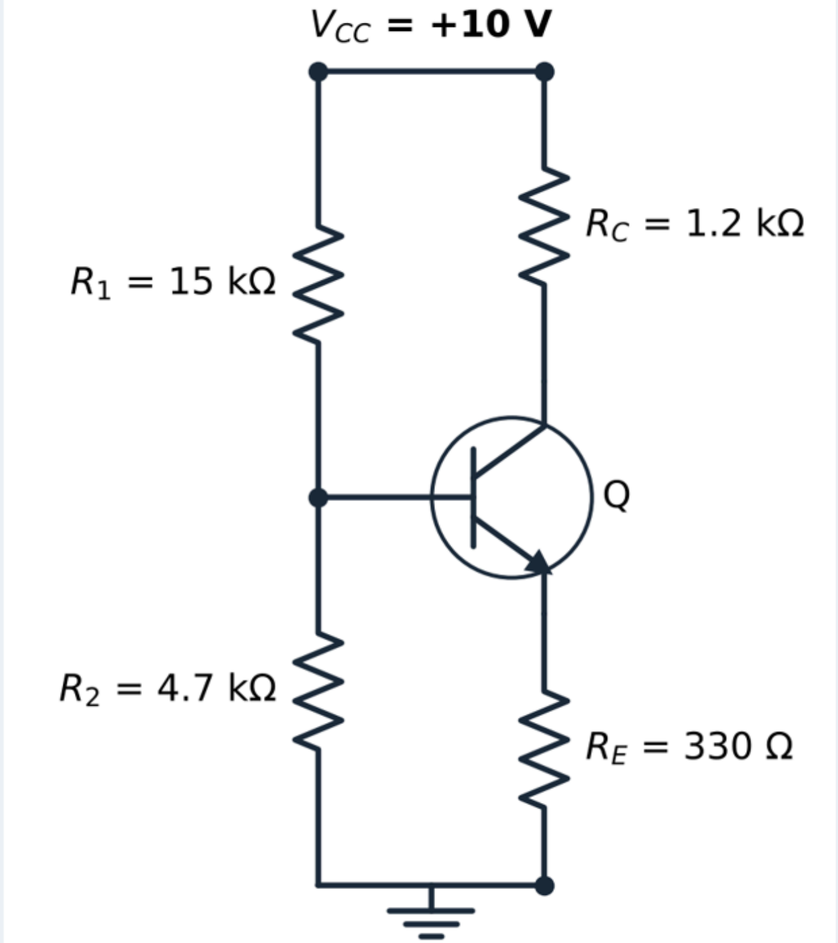
$\beta = 100$, $V_{BE} = 0.7\ V$, $V_{CE(sat)} = 0.2\ V$

Find

1. Find V_{TH} and R_{TH} of the divider.
2. Find I_B using $(\beta + 1)\cdot R_E$ in the base loop.
3. Find I_C, I_E and V_{CE} by hand.
4. Draw the load line (slope set by $R_C + R_E$).
5. Repeat with $\beta = 200$: how much does I_C change?


14
12
10
8
6
4
2
0
Collector current IC (mA)
0
2
4
6
8
10
12
14
Collector–emitter voltage VCE (V)

Class Assignment



Given

$V_{CC} = 10$ V
$R_1 = 15$ kΩ, $R_2 = 4.7$ kΩ
$R_C = 1.2$ kΩ, $R_E = 330$ Ω
$\beta = 100$, $V_{BE} = 0.7$ V, $V_{CE(sat)} = 0.2$ V

Find

1. Find V_{TH}, R_{TH} and I_B.
2. Find I_C, I_E, V_{CE}; draw the load line.
3. Power dissipated at the Q-point?
4. $\beta = 200$: % change in I_C?
5. Pick a standard R_C that gives $V_{CE} \approx 5$ V.

Class Assignment

